

Example: Estimating the effect of job training grants on scrap rates using panel data

In this example we set out to investigate if job training grants can reduce firm scrap rates. We have data from a Michigan job training program consisting of $N = 54$ firms over $T = 3$ years: 1987-1989. No firms received grants in 1987, 19 received grants in 1988, and 10 in 1989. A firm could not receive a grant twice. We have the following data on these firms over the three years (stored in `jtrain.dta`)

We write up our equation of interest

$$lscrap_{it} = \beta_0 + \beta_1 d88_{it} + \beta_2 d89_{it} + \beta_3 union_{it} + \beta_4 grant_{it} + \beta_5 grant_{i,t-1} + c_i + u_{it}$$

where the unobserved effect c_i captures e.g. worker and managerial ability. We include year dummies to control for the fact that workers might become more productive over time. Since there are firms in 1989 that received a grant only in 1988, it is important to allow the grant effect to persist one period (job training in 1988 could make workers more productive in 1989). Job training grants were approved on a first-come, first-serve basis, so we are concerned that the unobserved effect might be correlated with *grant*.

Variable	Description
$lscrap_{it}$	Log of the scrap rate for firm i at time t
$grant_{it}$	Dummy variable, equal to 1 if firm i received a grant at time t
$grant_{i,t-1}$	Dummy variable, equal to 1 if firm i received a grant at time $t-1$
$union_{it}$	Dummy variable, equal to 1 if firm i was a member of a union at time t
$d88_{it}$	Dummy variable, equal to 1 if year 1988
$d89_{it}$	Dummy variable, equal to 1 if year 1989
$fcode_{it}$	Identifier of firm i
$year_{it}$	

Important commands in Stata. Assume we have dependent variable (Y) and regressors X1, X2 and X3

- **Defining time and individual variable**
 - `xtset id time`
- **Estimating panel data model using Fixed Effects**
 - `xtreg Y X1 X2 X3, fe`
- **Estimating panel data model using Random Effects**
 - `xtreg Y X1 X2 X3, re`
- **Hausmann Test**
 - `hausman FErgress RErgress, sigmamore`
- **Estimating Panel Data with IV**
 - See `xtivreg` function

1- Pooled OLS

Source	SS	df	MS	Number of obs =	162
Model	16.5573816	5	3.31147633	F(5, 156) =	1.52
Residual	339.187701	156	2.17428013	Prob > F =	0.1856
				R-squared =	0.0465
				Adj R-squared =	0.0160
Total	355.745082	161	2.20959678	Root MSE =	1.4745

LSCRAP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D88	-.2412275	.3071864	-0.79	0.433	-.848009	.365554
D89	-.4723571	.334112	-1.41	0.159	-1.132324	.1876101
GRANT	.2052977	.3342902	0.61	0.540	-.4550215	.8656169
GRANT_1	-.0225261	.4321433	-0.05	0.958	-.8761333	.8310811
UNION	.5391657	.2465373	2.19	0.030	.0521836	1.026148
_cons	.4177121	.216836	1.93	0.056	-.0106013	.8460256

2- Fixed Effects

Fixed-effects (within) regression

Group variable: FCODE

Number of obs = 162

Number of groups = 54

R-sq: within = 0.2010

between = 0.0079

overall = 0.0068

Obs per group: min = 3

avg = 3.0

max = 3

F(4,104) = 6.54

corr(u_i, Xb) = -0.0714

Prob > F = 0.0001

LSCRAP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D88	-.0802157	.1094751	-0.73	0.465	-.297309	.1368776
D89	-.2472028	.1332183	-1.86	0.066	-.5113797	.0169741
GRANT	-.2523149	.150629	-1.68	0.097	-.5510178	.0463881
GRANT_1	-.4215895	.2102	-2.01	0.047	-.8384239	-.0047551
_cons	.5974341	.0677344	8.82	0.000	.4631142	.7317539
sigma_u	1.438982					
sigma_e	.4977442					
rho	.89313867	(fraction of variance due to u_i)				

F test that all u_i=0:

F(53, 104) = 24.66

Prob > F = 0.0000

3- Fist Difference

Source	SS	df	MS	Number of obs =	108
Model	6.59219195	4	1.64804799	F(4, 104) =	4.96
Residual	34.5904866	104	.332600833	Prob > F =	0.0011
				R-squared =	0.1601
				Adj R-squared =	0.1278
Total	41.1826786	108	.381321098	Root MSE =	.57672

D.LSCRAP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D88						
D1.	-.0906072	.0909695	-1.00	0.322	-.2710032	.0897888
D89						
D1.	-.2774225	.150368	-1.84	0.068	-.5756079	.0207628
GRANT						
D1.	-.222781	.1307423	-1.70	0.091	-.482048	.0364859
GRANT_1						
D1.	-.3512459	.2350848	-1.49	0.138	-.817428	.1149362

4- Fixed Effects vs First Difference?

Source	SS	df	MS	Number of obs	=	54
Model	.971328707	1	.971328707	F(1, 52)	=	3.10
Residual	16.312517	52	.31370225	Prob > F	=	0.0844
				R-squared	=	0.0562
				Adj R-squared	=	0.0380
Total	17.2838457	53	.326110296	Root MSE	=	.56009

err_fd	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
err_fd L1.	.2369064	.1346333	1.76	0.084	-.0332551	.5070678
_cons	1.31e-09	.0762188	0.00	1.000	-.1529442	.1529442

```
.      test L.err_fd = -1/2
```

```
( 1)  L.err_fd = -.5
```

```
      F( 1, 52) = 29.96
      Prob > F = 0.0000
```

```
.      test L.err_fd = 0
```

```
( 1)  L.err_fd = 0
```

```
      F( 1, 52) = 3.10
      Prob > F = 0.0844
```

5- Random Effects

Random-effects GLS regression

Group variable: FCODE

Number of obs = 162

Number of groups = 54

R-sq: within = 0.2006

between = 0.0206

overall = 0.0361

Obs per group: min = 3

avg = 3.0

max = 3

corr(u_i, X) = 0 (assumed)

Wald chi2(5) = 26.99

Prob > chi2 = 0.0001

LSCRAP	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D88	-.0934519	.1091559	-0.86	0.392	-.3073936	.1204898
D89	-.2698336	.1316496	-2.05	0.040	-.527862	-.0118052
GRANT	-.214696	.1477838	-1.45	0.146	-.504347	.0749549
GRANT_1	-.3770698	.2053516	-1.84	0.066	-.7795515	.0254119
UNION	.5478021	.410625	1.33	0.182	-.2570081	1.352612
_cons	.4148334	.2434322	1.70	0.088	-.062285	.8919517
sigma_u	1.3900287					
sigma_e	.4977442					
rho	.88634984	(fraction of variance due to u_i)				

6- Hausmann Test

	—— Coefficients ——		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) FErgress	(B) RErgress		
D88	-.0802157	-.0934519	.0132363	.0107669
D89	-.2472028	-.2698336	.0226308	.0219962
GRANT	-.2523149	-.214696	-.0376188	.0306006
GRANT_1	-.4215895	-.3770698	-.0445197	.0467426

b = consistent under Ho and Ha; obtained from xtreg

B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

```

chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        =          2.59
Prob>chi2 =          0.2738
(V_b-V_B is not positive definite)

```